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TECHNICAL NOTE

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TSDP: Automatic preprocessing and downloading of Landsat 1–9, Sentinel-2, and other remote sensing data for multi-resolution time-series image analysis

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ABSTRACT

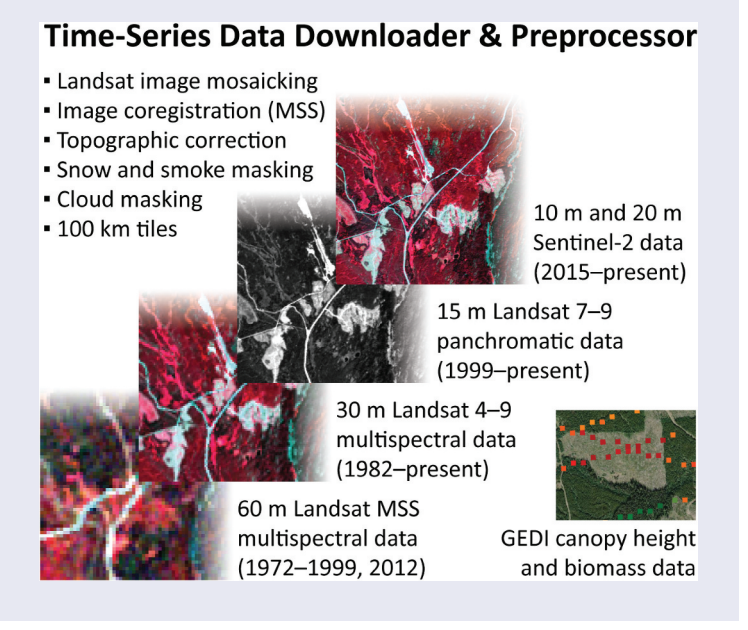
The Time-Series Data Downloader and Preprocessor (TSDP) is an open-source Python tool designed to streamline the acquisition and preprocessing of multispectral satellite imagery for time-series analysis. Built on Google Earth Engine (GEE), TSDP supports Landsat 1–9—from Landsat MSS to Landsat OLI—as well as Sentinel-2 imagery, GEDI canopy height and biomass products, and the recently released Google Satellite Embedding dataset. TSDP applies a consistent set of preprocessing routines across all optical sensors—including automatic image filtering, cloud, snow, and smoke masking, and topographic correction—generating analysis-ready time series spanning 1972 to the present. The tool integrates recent advances such as the Cloud Score+ algorithm for improved cloud detection in Sentinel-2 imagery and implements the SCS +C method for topographic correction using user-selectable digital elevation models (DEMs). TSDP prioritizes data quality by downloading imagery at its native spatial resolution and, in most cases, avoiding spatial resampling. This approach makes it particularly well suited for developing multi-resolution and multisource methods—an emerging direction in remote sensing for long-term environmental monitoring, land-cover change detection, and vegetation dynamics research. This technical note describes the tool's functionalities, input parameters, processing workflow, and outputs, providing guidance for users who wish to generate high-quality time series datasets from publicly available satellite archives.

KEYWORDS

Python; Sentinel-2; Google Satellite Embedding; panchromatic; multi-sensor; MSS

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1. Introduction

Geospatial cloud-computing platforms, particularly Google Earth Engine (GEE), have transformed remote sensing by enabling large-scale data processing without local infrastructure (Alshehri et al., 2025; Pham-Duc et al., 2023). However, while GEE excels at pixel-based time-series algorithms like LandTrendr (Landsat-based detection of Trends in Disturbance and Recovery; Kennedy et al., 2018) and CCDC (Continuous Change Detection and Classification; Gorelick et al., 2023), it is less suited for tile-based techniques that operate on entire scenes (Parshakov et al., 2025). More fundamentally, the platform's memory and compute constraints, reliance on built-in routines, and restricted access to low-level data manipulations limit its flexibility for custom algorithm development (Alshehri et al., 2025). In addition, despite the increasing adoption of deep learning within the remote sensing community, GEE offers limited support for such workflows: model training remains computationally infeasible within the platform and must be performed externally (Alshehri et al., 2025; Yang et al., 2022; Zhang et al., 2022). Therefore, offline processing remains essential for developing novel information extraction techniques, offering researchers greater control over algorithms, input data, and intermediate products.

Data quality and availability can also favor local data management. For example, the European Space Agency (ESA) periodically reprocesses the entire Sentinel-2 archive (2015–present) with improved algorithms and makes all reprocessed data available through the Copernicus Data Space Ecosystem, providing the most recent processing baseline (Enache et al., 2023). Based on our inspection of image metadata at the time of writing (February 2026), Sentinel-2 images hosted on GEE appear to remain at the processing baseline available at ingestion and are not retroactively updated. However, because GEE provides improved cloud masks for Sentinel-2 (Pasquarella et al., 2023),

accessing both up-to-date processing baselines and advanced masks requires downloading imagery from Copernicus and cloud masks from GEE separately.

Constructing high-quality image time series can be challenging, as even a single location may require hundreds of scenes spanning multiple decades and sensors. Manually downloading such volumes of data from multiple sources is time-consuming. Although batch-download tools exist, they still require substantial effort, including subsetting imagery, organizing files, and performing additional preprocessing.

Several tools aim to automate these tasks and reduce the manual burden of satellite data acquisition. For example, GEE Dimensional Image Manager (geedim; Gedam, 2025) automatically selects Landsat 4–9 and Sentinel-2 data based on cloud cover, applies cloud masking (including improved cloud masking for Sentinel-2; Pasquarella et al., 2023), clips the imagery to the user's area of interest (AOI), and downloads the resulting time series from GEE. The Semi-Automatic Classification Plugin (SCP; Congedo, 2021) integrates with QGIS to offer a convenient interface for selecting areas of interest on a map, downloading Landsat and Sentinel-2 imagery from multiple platforms with basic cloud masking based on quality bands or metadata, while also providing tools for data processing (e.g., image classification). Earth Observation Data Access Gateway (EODAG; CS SI, 2025) supports downloading from a particularly wide array of data providers—Copernicus, USGS, AWS, Microsoft's Planetary Computer, and several others—and filters images by cloud cover, but typically returns full scenes without subsetting or tiling. Finally, the Time Series Downloader (TSD; de Franchis, 2024) downloads Sentinel, Landsat, and Planet image crops of configurable size (e.g., $5{,}000 \times 5{,}000$ m) centered at a point with user-defined coordinates, with basic cloud masking.

While these tools facilitate various aspects of preparing time-series satellite datasets for offline analysis, they share several common limitations. They do not support Landsat MSS, lack integrated topographic correction or specialized masks (e.g., snow and smoke), and several do not implement advanced cloud masking methods for Sentinel-2. Compounding these issues, they offer no solution for inconsistent preprocessing across sensors—for instance, Sentinel-2 surface reflectance (SR) data from ESA includes topographic correction, whereas Landsat SR data does not. Consequently, a clear gap remains for users seeking consistently preprocessed, analysis-ready image time series spanning 1972 to the present.

Existing analysis-ready products like Harmonized Landsat Sentinel-2 (HLS) and Landsat ARD address some preprocessing needs but impose significant tradeoffs (Claverie et al., 2018; Dwyer et al., 2018; Ju et al., 2025). The convenience of these products stems from their consistent tiling: each tile shares a common projection, pixel grid, spectral configuration, and preprocessing across sensors, allowing straightforward bulk download and immediate use. However, Landsat ARD is limited to the United States, while HLS covers only 2013–present and resamples data to a common 30 m pixel grid (distinct from the original Landsat grid), reducing spatial detail in both Sentinel-2 (due to resampling from 10 m and 20 m to 30 m) and Landsat (due to pixel grid shift and associated interpolation) data (Claverie et al., 2018; Ju et al., 2025). While these tradeoffs accommodate traditional pixel-based analyses requiring strict cross-sensor uniformity, they diminish information content that emerging multi-resolution methods could exploit.

We introduce the Time-Series Data Downloader and Preprocessor (TSDP), a Python tool that generates analysis-ready time-series datasets specifically for multi-sensor,

multi-resolution information extraction. TSDP supports all Landsat sensors (1–9), Sentinel-2, GEDI, and other products, organizing data by Military Grid Reference System (MGRS) tiles for consistent spatial coverage. Unlike harmonized products, TSDP retains sensor-specific characteristics—including 10 m Sentinel-2 bands and 15 m Landsat panchromatic data—preserving the spatial detail critical for applications like detecting narrow logging roads or small forest clearings. TSDP integrates comprehensive preprocessing within a single reproducible workflow: automated acquisition from multiple platforms (GEE, Copernicus, NASA Earthdata), topographic correction, coregistration, and masking of clouds, snow, and smoke. This enables researchers to focus on developing innovative time-series analysis techniques rather than data preparation.

This manuscript provides an overview of the main capabilities and configurable parameters of TSDP, along with the general workflow for preparing and executing a download. We outline how the algorithm interfaces with GEE and other data providers—such as the Copernicus Data Space Ecosystem and NASA's Earthdata platform—to retrieve time-series datasets according to user-defined specifications. The paper describes the functionality, scope, and application context of TSDP, while additional low-level technical details—such as explanations of less critical parameters and embedded code comments—are provided in the TSDP algorithm itself, implemented as a Python Jupyter Notebook (an interactive Python script format) and available on GitHub: https://github.com/parshakov-code/time-series-data-downloader.

2. Data

The algorithm downloads data from various sources, including GEE, the Copernicus Data Space Ecosystem, and NASA Earthdata. The acquired data can include raw digital number (DN), top-of-atmosphere (TOA) reflectance, and surface reflectance (SR) products from Landsat and Sentinel-2, as well spaceborne LiDAR data (GEDI) and auxiliary data for preprocessing.

2.1. Optical satellite data

The algorithm can download data from Landsat 1–5 MSS, Landsat 4–5 TM, Landsat 7 ETM +, Landsat 8–9 OLI, and Sentinel-2 MSI. The characteristics of these sensors are summarized in Table 1. Landsat MSS and Landsat 7–9 panchromatic data are only available as digital numbers (DN), representing scaled, calibrated at-sensor radiance. For multispectral data from Landsat 4 TM and newer 30 m Landsat sensors, users can choose between raw DN and SR products. Sentinel-2 imagery is available as top-of-atmosphere (TOA) or SR products. The user can choose what spectral bands to download.

Landsat data are obtained from GEE (Collection 2) and/or NASA Earthdata (HLS), while Sentinel-2 data are downloaded from GEE, Copernicus Data Space Ecosystem, and/or NASA Earthdata (HLS). The bit depth is 8-bit for MSS, TM, and ETM+ raw DN data without topographic and atmospheric correction and 16-bit for all other preprocessing levels and sensors. Notably, 8-bit data require half the storage of 16-bit data, which benefits applications that do not require atmospheric correction, such as UC-Change. Storage requirements are discussed further in Section 4.1.

Table 1. The characteristics of multispectral data downloaded by TSDP. Note that signal-to-noise ratio (SNR) depends on the at-sensor radiance and that SNR for different sensors was measured at different radiance levels, usually representing typical conditions over land (Gascon et al., 2017; Mika, 1997; Morfitt et al., 2015; Okin & Gu, 2015). The noise equivalent change in temperature (NEΔT) is used to characterize Thermal band noise performance instead of singal-to-noise ratio (SNR). For example, Landsat 8 was reported to have four times better NEΔT than Landsat 7 (Haque et al., 2023).

Sensors, their radiometric resolution and service years	Spectral Band	Description	Wavelength Range (μm)	Spatial Resolution (m)	SNR
Landsat 1–5 MSS (6 bit)	Band 1	Green	0.5–0.6	68 × 83	–
July 1972 to October 1999, June 2012 to January 2013	Band 2	Red	0.6–0.7	68 × 83	–
	Band 3	NIR 1	0.7–0.8	68 × 83	–
	Band 4	NIR 2	0.8–1.1	68 × 83	–
Landsat 4–5 TM (8 bit)	Band 1	Blue	0.45–0.52	30	32
July 1982 to May 2012	Band 2	Green	0.52–0.60	30	17
	Band 3	Red	0.63–0.69	30	18
	Band 4	NIR	0.76–0.90	30	15
	Band 5	SWIR 1	1.55–1.75	30	26
	Band 6	Thermal	10.40–12.50	120	–
	Band 7	SWIR 2	2.08–2.35	30	18
Landsat 7 ETM+ (8 bit)	Band 1	Blue	0.45–0.52	30	39
July 1999 to January 2024	Band 2	Green	0.52–0.60	30	37
	Band 3	Red	0.63–0.69	30	26
	Band 4	NIR	0.77–0.90	30	34
	Band 5	SWIR 1	1.55–1.75	30	36
	Band 6	Thermal	10.40–12.50	120	–
	Band 7	SWIR 2	2.09–2.35	30	27
	Band 8	Pan	0.52–0.90	15	16
Landsat 8 (12 bit) and 9 (14 bit) OLI/TIRS	Band 2	Blue	0.45–0.51	30	367
April 2013 to present	Band 3	Green	0.53–0.59	30	304
	Band 4	Red	0.64–0.67	30	227
	Band 5	NIR	0.85–0.88	30	201
	Band 6	SWIR 1	1.57–1.65	30	267
	Band 7	SWIR 2	2.11–2.29	30	327
	Band 8	Pan	0.50–0.68	15	148
	Band 10	Thermal 1	10.60–11.19	100	–
	Band 11	Thermal 2	11.50–12.51	100	–
Sentinel-2 (12 bit)	Band 2	Blue	0.46–0.53	10	214
June 2015 to present	Band 3	Green	0.54–0.58	10	249
	Band 4	Red	0.65–0.68	10	230
	Band 5	Red edge 1	0.78–0.89	20	253
	Band 6	Red edge 2	0.70–0.71	20	220
	Band 7	Red edge 3	0.73–0.75	20	227
	Band 8	NIR	0.77–0.79	10	221
	Band 8A	Narrow NIR	0.85–0.87	20	161
	Band 11	SWIR 1	1.57–1.66	20	158
	Band 12	SWIR 2	2.10–2.28	20	166

Images are downloaded at their original spectral and spatial resolutions for all sensors except for Landsat MSS and the thermal bands of Landsat 4 TM–9 OLI. The original 68 m × 83 m resolution of Landsat MSS is resampled to 60 m by the data provider, United States Geological Survey (USGS), while the thermal bands of Landsat 4 TM–9 OLI are resampled to 30 m by USGS. In addition, some Landsat MSS images are missing specific spectral bands. For instance, all Landsat MSS scenes from 2012 to 2013 lack valid data in the NIR2 band (USGS, n.d.); therefore, TSDP downloads only the remaining three bands (Green, Red, and NIR1) for those images.

It is also important to note that Landsat and Sentinel-2 pixel grids do not align due to differences in how pixel coordinates are defined. In Landsat data, pixel coordinates refer to the center of the pixel, whereas in Sentinel-2 data, they correspond to the

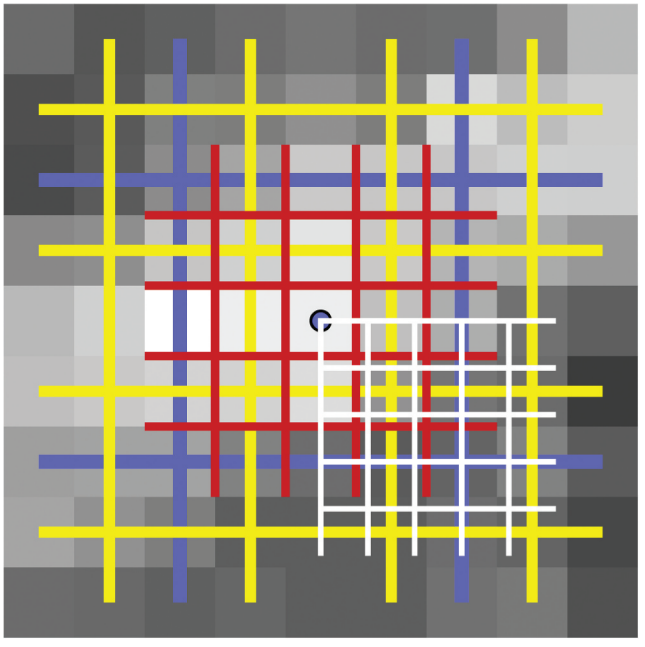


Figure 1. This image illustrates the pixel grids of Landsat MSS (blue), 30 m Landsat 4 TM–9 OLI (yellow), 15 m Landsat 7–9 panchromatic (red), and Sentinel-2 (white) data overlaid on top of a Landsat 7 panchromatic image. The blue dot in the center represents the matching geographic coordinates of the pixels.

northeast (NE) corner of the pixel (Figure 1). This difference introduces a systematic spatial shift when comparing or combining images from both sensors, even when they share the same coordinate system (i.e., the same UTM zone). This misalignment presents a challenge for multi-sensor image analysis. For example, although the spatial resolutions of the Landsat panchromatic band (15 m), 30 m Landsat 4 TM–9 OLI multispectral bands, and 60 m Landsat MSS bands are multiples of each other, their pixel grids do not align. As a result, simply averaging four adjacent 15 m panchromatic pixels to produce an image with 30 m resolution will create a grid that does not align with the 30 m multispectral bands due to a 7.5 m shift in both the northing and easting directions. TSDP preserves the original pixel grids of Landsat Collection 2 and Sentinel-2 Level 1C and 2A data, and researchers using the TSDP algorithm to download data must take this into account when developing multi-resolution and multi-sensor techniques.

In contrast, the pixel grids of Sentinel-2's 10 m, 20 m, and 60 m bands align perfectly. For example, averaging four adjacent 10 m VNIR pixels produces a 20 m image that is perfectly aligned with the 20 m red-edge and SWIR bands, ensuring seamless integration across Sentinel-2 spectral bands.

In HLS data, harmonized Landsat and Sentinel-2 images use the same 30 m pixel grid. To achieve this, NASA spatially resamples images from both sensors. For example, 30 m Collection 2 Landsat bands are resampled to a pixel grid matching Sentinel-2 data, requiring half a pixel (15 m) shift in both the northing and easting directions (Claverie

et al., 2018). TSDP offers the option to download HLS data for users who require imagery with consistent spatial resolution.

2.2. Auxiliary data for preprocessing

TSDP allows users to define a temporal window for downloading satellite imagery by specifying a custom range of years and days of the year (DOY). Alternatively, users can opt to restrict downloads to the full growing season or to its peak period (mid-summer). The start and end of the growing season are automatically derived from MODIS land surface phenology metrics provided by the global MCD12Q2 dataset (Gray et al., 2019). Specifically, TSDP uses the MidGreenup_1 metric (midpoint of spring green-up, indicating that active growth has begun) and the MidGreendown_1 metric (midpoint of greendown, marking the start of widespread senescence). To focus on the peak of the growing season—typically corresponding to mid-summer—TSDP uses the Maturity_1 metric (the time when vegetation reaches its maximum photosynthetic activity) and the Senescence_1 metric (beginning of decline from peak condition). Together, the two metrics define a narrower window centered on maximum vegetation productivity, enabling more targeted analyses during the most ecologically active period. Currently, the tool does not account for within-tile spatial variation in growing season length (e.g., shorter seasons in high-elevation areas); instead, phenology metrics are averaged across the entire tile.

The user can choose to mask water bodies and/or agricultural lands as such areas may trigger false detections of change by change detection algorithms. The corresponding masks are derived from the 10 m ESA WorldCover map, available for the years 2020 and 2021. For Canada, the user can choose to use the Agriculture and Agri-Food Canada (AAFC) map for higher accuracy (Fisette et al., 2013; Venter et al., 2022). Annual AAFC maps are available for 2009 to present for the prairie provinces of Canada (Alberta, Manitoba, and Saskatchewan), and 2011 to present for most other provinces.

GEE's Cloud Score+ data is used to mask clouds and cloud shadows in Sentinel-2 data with much higher accuracy than using the quality band provided with the Sentinel-2 imagery (Pasquarella et al., 2023). In contrast, the quality band provided with Landsat data provides high-quality cloud masking, and therefore no additional data are required to mask clouds in Landsat images.

For topographic correction, users can select from several Digital Elevation Models (DEMs) available on GEE, including NASADEM and the Copernicus DEM (GLO-30). NASADEM is an enhanced version of the original Shuttle Radar Topography Mission (SRTM) dataset, incorporating auxiliary data from sources such as ASTER GDEM, ICESat GLAS, and PRISM to improve its quality and consistency (Simard et al., 2024). In contrast, GLO-30 is strictly a digital surface model (DSM) derived from TanDEM-X data. Unlike a terrain model, it represents the Earth's surface including buildings, vegetation, and other vertical structures.

This distinction can pose challenges when analyzing dynamic landscapes like managed forests, where above-ground features such as trees or infrastructure may change over time. Since GLO-30 was generated from data collected during a specific period, it may still depict features—such as trees in areas that were later clearcut—that no longer exist.

Although both datasets are provided at a 30 m resolution, GLO-30 typically exhibits finer spatial detail due to the higher native resolution of the TanDEM-X sensor (~12 m) compared to the ~30 m resolution of SRTM. Several studies have found that GLO-30 offers better vertical accuracy and is less sensitive to terrain slope (Meadows et al., 2024; Simard et al., 2024). However, because it captures more of the vegetation canopy, it may deviate more from true ground elevation in forested regions. NASADEM, while also not a true Digital Terrain Model (DTM), is less influenced by above-ground features and more closely approximates bare-earth conditions. For instance, clearcut areas are far less apparent in NASADEM compared to GLO-30. This is likely due to differences in radar wavelength: the X-band radar used in TanDEM-X penetrates vegetation less effectively than the longer-wavelength C-band radar used in SRTM.

2.3. GEDI data

GEDI-derived canopy height and biomass data available for download from GEE include GEDI L2A Raster Canopy Top Height and GEDI L4A Raster Aboveground Biomass Density products. To facilitate the interpolation of GEDI-derived canopy height and biomass estimates using optical data, TSDP resamples these raster products from their 25 m resolution to match the spatial resolution of the chosen optical sensors. Canopy height values are converted from floating-point meters to 16-bit integers in centimeters to reduce storage requirements while minimally affecting measurement precision.

To ensure reliability, TSDP applies the quality masks provided with the GEDI products. Specifically, it keeps only observations that (i) pass the Level-4 quality check, (ii) are not flagged as degraded, and (iii) were collected under leaf-on conditions. After masking, only the relevant science variables—including Relative Height (RH) metrics, Aboveground Biomass Density (AGBD), and the day of month of the corresponding measurements—are retained for analysis.

As with other satellite sensors, TSDP downloads monthly GEDI data either as NumPy arrays, GeoTIFFs, or both. When saved as NumPy arrays, TSDP separates the information into two files per month. The first file is a 2D "date mask" (e.g., 2020-12-01_GD_date_20m.npy for December 2020), where non-zero pixels mark GEDI shot locations, and the pixel values indicate the day of the month (DOM) when the measurements were collected. The second file (e.g., 2020-12-01_GD_data_20m.npy) contains the actual measurement data-four bands corresponding to RH50, RH70, RH95, and AGBD—for the same locations defined in the 8-bit date mask. This structure acts as a highly efficient form of lossless compression, linking each GEDI measurement to its acquisition date while enabling seamless handling of canopy height and biomass data across large areas. In contrast, GeoTIFFs store the same information directly as five bands: RH50, RH70, RH95, AGBD, and DOM.

2.4. Google Satellite Embedding dataset

The Google Satellite Embedding dataset is a new global product generated by the AlphaEarth Foundations model, a geospatial embedding model developed by Google (Brown et al., 2025). AlphaEarth integrates information from multiple satellite datastreams —including optical, radar, and LiDAR—into a common feature space that represents

surface conditions. Instead of storing raw imagery, the dataset provides compact numerical embeddings that capture spectral, structural, and environmental patterns. These embeddings can be used for tasks such as land-cover classification, change detection, and ecological monitoring.

The embeddings are originally provided as 64-band images in double precision values ranging from −1.0 to 1.0. At 10 m resolution, file sizes are very large—approximately 54 GB per year per MGRS tile. To reduce storage requirements and simplify processing, TSDP converts these values to unsigned 16-bit integers (ranging from 1 to 65,535), with 0 reserved for masked pixels (e.g., water bodies or agricultural areas). As with GEDI data, TSDP also provides the option to resample the embeddings to match the spatial resolution of the chosen optical sensors, making the data more manageable—e.g., 1.5 GB per year per tile at 30 m resolution. Spatial resampling is performed using the pixel aggregation method, in which pixel values are computed as area-weighted averages based on the fractional overlap between the finer-resolution input pixels and the coarser pixels defined by the output projection (Google, 2024).

2.5. Change detection and other maps

TSDP can download the global and regional forest change detection maps available on GEE, including the Global Forest Change (GFC) and Continuous Change Detection and Classification (CCDC) maps. Developed by Hansen et al. (2013), the GFC dataset offers global maps of forest cover change at a 30-meter resolution. It provides information on forest extent, loss, and gain from 2000 through 2024 (a new version is released every year), derived from time-series analysis of Landsat images. The CCDC dataset contains precomputed results from applying the Continuous Change Detection and Classification algorithm to 20 years of Landsat SR data, spanning from January 1, 1999, to January 1, 2020.

TSDP can also download subsets of the 30 m forest-canopy height map developed by Potapov et al. (2021). This map integrates GEDI LiDAR RH95 metrics collected between April and October 2019 with the 2019 Landsat time series. A locally calibrated, moving-window ensemble of bagged regression trees was then used to extrapolate footprint-level GEDI measurements across latitudes 52° N to 52° S.

Another canopy-height map available for downloading by TSDP is the global 10 m map produced by Lang et al. (2023). This dataset was generated using a deep learning model trained on over 2.7 million GEDI footprints and high-resolution Sentinel-2 imagery. The approach leverages a convolutional neural network to learn spatial patterns between optical reflectance and canopy height, enabling prediction of canopy structure at fine spatial resolution across the globe.

3. TSDP algorithm description and parameters

TSDP is implemented as a Python script presented in the form of a Jupyter Notebook. It can be opened and executed using any compatible code editor that supports Jupyter environments, such as Microsoft Visual Studio Code. Upon first execution, the script automatically installs all necessary Python packages. However, before running the script, users must ensure that they have a GEE Cloud Project associated with their Google

account. If not already set up, this can be done by following the official GEE guidelines for creating a Cloud Project (Google, 2025).

The current implementation of the script does not feature a graphical user interface. Instead, all user-defined parameters must be entered manually within one of the initial cells of the Jupyter Notebook. The following subsections provide a detailed explanation of each input parameter required by the TSDP algorithm, which are summarized in Table 2.

3.1. Copernicus Data Space Ecosystem login credentials

If the user intends to download consistently preprocessed Sentinel-2 data, they must provide their username and password for the Copernicus Data Space Ecosystem. This is particularly important for accessing Sentinel-2 imagery processed with the most recent preprocessing algorithms. As discussed in the Introduction, GEE does not retroactively update its archive with reprocessed Sentinel-2 images. Therefore, to ensure consistency in atmospheric, topographic, and geometric corrections across the time series, especially for change detection studies, it is recommended to download data directly from the Copernicus repository instead of GEE.

3.2. NASA Earthdata login credentials

The parameters username_nasa and password_nasa refer to the user's login credentials for NASA's Earthdata portal (https://www.earthdata.nasa.gov/). These are required only when downloading Harmonized Landsat and Sentinel-2 (HLS) SR products at 30 m resolution. Users who do not intend to use HLS data may leave these fields blank.

3.3. GEE access and initialization

To access GEE resources, users must first authenticate and initialize their connection to the GEE platform. This process establishes a secure link between the script and the user's GEE account, enabling access to Earth observation datasets and cloud-based processing services. During the first run of the TSDP script, the authentication step opens a browser window directing the user to a Google login page, where they are prompted to generate an authorization token. After selecting the Google account linked to GEE, users must grant permissions to view and manage Earth Engine resources and Google Cloud Storage data.

Although a warning may appear stating that "Google hasn't verified this app," this message is standard for open-source applications using the Earth Engine API and can be safely ignored if the script is trusted. TSDP is fully open source, and users are encouraged to inspect the code for transparency. The script does not collect or transmit any personal information beyond what is required for accessing satellite data. User credentials for external data portals—such as NASA Earthdata and the Copernicus Data Space—are transmitted securely and directly to their respective authentication services.

Once access is granted, the user copies the generated code back into their development environment (e.g., Visual Studio Code) to complete the authentication. The credentials are then stored locally, allowing future sessions to run without re-authentication—unless a prolonged period of inactivity occurs (typically around 2 weeks).

Table 2. Configuration parameters for TSDP tool by sensor type.

Parameters and their default values	Description
Common Parameters (All Sensors)	
username_esa and password_esa	User's login credentials for the Copernicus Data Space Ecosystem—required for downloading Sentinel-2 data from the Copernicus Hub.
username_nasa and password_nasa	User's login credentials for NASA's Earthdata portal—required only for downloading HLS SR products.
MGRS_tile = '10UCV'	The MGRS tile covering the user's study area.
tile_overlap_m = 2960	Defines the buffer (in meters) by which each MGRS tile is extended eastward and southward to ensure spatial overlap between adjacent tiles (maximum 9800 m—the original Sentinel-2 tile buffer).
start_year = 1972	Specifies the initial year of the analysis period.
end_year = datetime.now().year	Specifies the final year of the analysis period.
start_DOY = 0	Start day-of-year for image acquisition; determined automatically using MODIS phenology data if end_DOY is set to 0.
end_DOY = 0	End day-of-year for image acquisition; determined automatically using MODIS phenology data if set to 0.
mid_summer_only = False	Download midsummer data only: use MCD12Q2 Maturity_1 for start_DOY instead of MidGreendown_1 and Senescence_1 for end_DOY instead of MidGreendown_1.
do_sensors = ['M', 'L', 'S2']	Specifies which sensor series to download. Users can select Landsat MSS ('M'), Landsat TM–OLI ('L'), and Sentinel-2 ('S2'), among other available options.
bands	Dictionary containing lists of bands to download for each sensor. Default: {'M': ['G', 'R', 'NIR1', 'NIR2'], 'L': ['B', 'G', 'R', 'NIR', 'SWIR1', 'SWIR2', 'T'], 'S2_10m': ['B', 'G', 'R', 'NIR_wide'], 'S2_20m': ['RE1', 'RE2', 'RE3', 'NIR', 'SWIR1', 'SWIR2'], 'S2': ['B', 'G', 'R', 'RE1', 'RE2', 'RE3', 'NIR', 'SWIR1', 'SWIR2']}
sr = False	If True, TSDP downloads SR data; otherwise, TOA/DN depending on availability. Note: USGS does not offer SR Landsat MSS data and Landsat 7–9 panchromatic data.
do_topo = False	Enables topographic illumination correction for all sensors, excluding Level 2A (SR) Sentinel-2 data, which is already topographically corrected.
mask_agriculture = True use_AAFC = True	If True, TSDP masks agricultural areas using the global ESA WorldCover 10 m map and, optionally, AAFC crop maps for Canada.
min_coverage = 0.95	Minimum fraction of tile area required to be covered annually by imagery.
cloud_buffer_m = 240	The width of buffer around clouds for MSS, L, and S2 (meters). Also applied to snow and wildfire smoke.
Landsat MSS Parameters	
coregister_MSS = False	If True, TSDP coregisters MSS images to a cloud-free Landsat 4/5 TM image or image composite.
convert_MSS_to_TOA = False	Raw DN to TOA reflectance conversion using metadata coefficients and the solar zenith angle.
MSS_bright_thres_NDGR = 0.0	Normalized Difference Green-Red index threshold for cloud detection.
MSS_bright_thres_green = 15	Green TOA reflectance (%) threshold for detecting clouds, snow, and smoke; used in combination with MSS_bright_thres_NDGR.
MSS_bright_thres_green_ultimate = 25	Green TOA reflectance (%) threshold that ignores MSS_bright_thres_NDGR.
MSS_dark_thres_NIR2 = 10	NIR2 TOA reflectance (%) threshold used for two purposes: (1) filtering out MSS images with incorrect georeferencing (see Section 3.4), and (2) initial detection of cloud shadows for filtering out MSS images with high cloud cover. The dark threshold for the final cloud shadow mask is determined automatically in subsequent steps.
Landsat TM–OLI Parameters	
min_pan_coverage = 0.9	Yearly tile coverage requirement for 1999–2011 Landsat 7 panchromatic imagery (min_coverage is used for other years and sensor series).
Landsat TM–OLI and Sentinel-2 Parameters	
do_hd = ['L', 'S2']	Controls whether higher-resolution spectral bands are downloaded at native resolution for selected sensors (15 m panchromatic for Landsat 7–9; 10 m VNIR for Sentinel-2). Including 'S2' in this list saves the 10 m bands both as a separate native-resolution copy and as part of a 20 m stack with other spectral bands.

(*Continued*)

Table 2. (Continued).

Parameters and their default values	Description
do_HLS = False	Controls whether to download the HLS version of Sentinel-2 and Landsat 8 and 9 data.
smoke_thres_aerosol_toa = 15 smoke_thres_aerosol_sr = 8	TOA or SR aerosol band threshold (%) for smoke detection in Level 1C or Level 2A Landsat 8 and 9 OLI data.
smoke_thres_blue_toa = 15 smoke_thres_blue_sr = 10	TOA or SR blue band threshold (%) for smoke detection in Level 1C or Level 2A Landsat 4–7 TM and ETM+ data.
smoke_thres_swir1_toa = 15 smoke_thres_swir1_sr = 18	TOA or SR SWIR1 band threshold (%) for smoke detection in Landsat TM–OLI data.
Sentinel-2 Parameters	
s2_cloud_mask_type = 'cs'	Selects Cloud Score+ mask type: 'cs' or 'cs_cdf'. "The cs band tends to be more sensitive to haze and cloud edges [. . .] while cs_cdf tends to be less sensitive to these low-magnitude spectral changes as well as terrain shadows" (V. Pasquarella, 2023)
s2_cloud_score_threshold = 0.8	Threshold for Sentinel-2 cloud masking using Cloud Score+ (typical range: 0.50–0.65; 0.8 for more aggressive masking).
snow_thres_green_toa = 20	Green TOA (%) threshold for snow detection.
snow_thres_swir1_toa = 15	SWIR1 TOA (%) threshold for snow detection.
ndsi_thres_s2 = 0.0	Normalized Difference Snow Index threshold for snow detection in Level 2A (SR) data. NDSI = 0.0 is a very aggressive threshold; however, it produces few false detections in forested areas. Typically recommended value for SR data is 0.4 (Section 3.8).
download_ESA_S2 = True	If True, TSDP downloads Sentinel-2 data from the Copernicus Data Space portal instead of GEE.
mask_esa_s2_with_gee = True	If True, TSDP applies GEE Cloud Score+ mask to Sentinel-2 imagery even if it is downloaded from the Copernicus website.
Other Sensors and Maps	
GEDI_resolutions = ['L', 'S2']	Original 25 m monthly rasterized GEDI data (GEDI L2A Canopy Top Height and GEDI L4A Aboveground Biomass Density products) are resampled to 20 or 30 m to match the pixel grids of Landsat TM-OLI, Sentinel-2, or any other optical sensor included in this list, using the nearest neighbour method.
embedding_resolutions = ['L']	Yearly Google Satellite Embedding composites are downloaded at the spatial resolution of optical sensors listed here.

3.4. Military Grid Reference System (MGRS) tile

TSDP organizes downloaded imagery by tile, using the same Military Grid Reference System (MGRS) employed by the Sentinel-2 mission. MGRS provides a standardized and globally consistent tiling scheme based on the Universal Transverse Mercator (UTM) projection, making it well suited for integrating data from multiple sensors with varying spatial and temporal characteristics. By adopting this tiling system, TSDP ensures compatibility with existing Sentinel-2 processing frameworks and simplifies multi-source data alignment.

To begin using TSDP, users must specify the MGRS tile corresponding to their study area. This can be done by consulting online tools such as GISSurfer (GISSurfer, n.d.), which allows users to identify tile codes using a map. Alternatively, users may refer to the Sentinel-2 Tiling Grid Parameters (TILPAR) KML file available from the Copernicus website (European Space Agency, n.d.). For example, to retrieve imagery for the area including Ottawa, Canada, the correct MGRS tile code is "18TVR."

In the Sentinel-2 tiling system, each MGRS tile covers a 100 km × 100 km area plus a 9.8 km spatial overlap with adjacent tiles, ensuring full coverage and minimizing edge artifacts during image processing. However, this overlap also results in substantial data redundancy and increased storage requirements (Bauer-Marschallinger & Falkner, 2023).

To address this, TSDP allows users to reduce the tile overlap by specifying a custom buffer width (tile_overlap_m) that suits their particular application. For instance, in clearcut detection, where features of interest are small, the overlap could be safely reduced to 1,000 meters or less. Conversely, for applications such as wildfire burn scar mapping, where large contiguous areas are important, a wider buffer (e.g., 3,000 meters) may be necessary to ensure full representation of the disturbance within at least one tile.

3.5. Landsat MSS image filtering and coregistration

Many Landsat MSS images are very noisy and unsuitable for most remote sensing applications (Figure 2). Additionally, the Landsat MSS archive currently available on GEE contains many scenes with incorrect georeferencing. Most remaining images have relatively low geometric accuracy compared to Landsat TM–OLI and Sentinel-2 imagery, and do not meet the USGS Tier 1 criterion, which requires a maximum of 12-meter radial root mean square error (RMSE).

To address these issues—namely, to filter out very low-quality images and those with incorrect coordinates, and to improve the geometric accuracy of the remaining images—TSDP uses a high-quality reference Landsat TM image. This reference image is the most cloud-free Landsat 4 or 5 TM image acquired between 1982 and 1993 that provides the best coverage of the tile. If the tile lies between two Landsat TM paths and no single image fully covers it, a composite of multiple images is used instead.

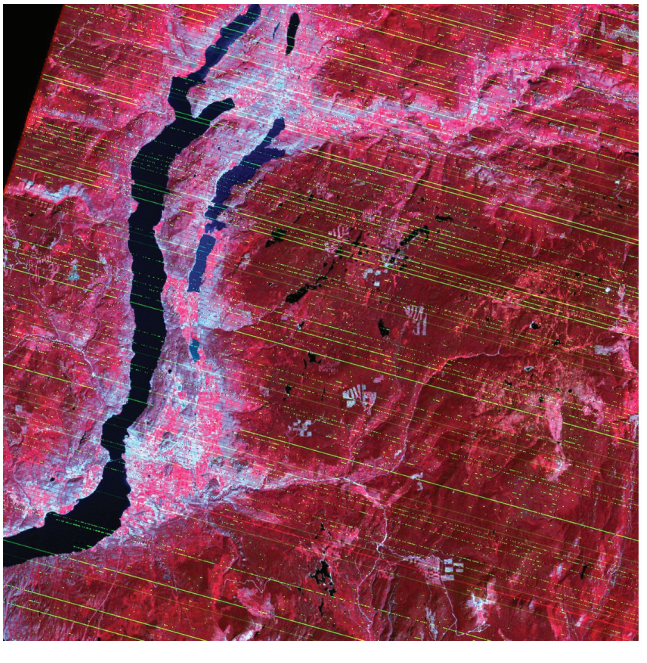


Figure 2. A low-quality Landsat 1 MSS image acquired on 1972-07-27 in tile 11ULR (104 km × 104 km; center coordinates: 50°3′N, 119°1′W; RGB = NIR2-red-green). This tile covers a mountainous, forested area around the city of Kelowna, British Columbia, Canada.

Table 3. Correlation matrix for a high-quality Landsat 1 MSS image acquired on 1973-08-10 in tile 11ULR (same tile as in Figure 2 but different image). Note the high green–red and NIR1–NIR2 correlations, as expected for a forested area.

	Band 1 (green)	Band 2 (red)	Band 3 (NIR 1)	Band 4 (NIR 2)
Band 1	1	0.97	0.72	0.57
Band 2	0.97	1	0.75	0.60
Band 3	0.72	0.75	1	0.97
Band 4	0.57	0.60	0.97	1

Images with inaccurate geographic referencing are detected by measuring the NIR2 (Band 4) TOA reflectance in pixels identified as water in the reference TM image. If the average reflectance of these pixels exceeds the MSS_dark_thres_NIR2, the image is rejected due to likely misregistration.

Noisy Landsat MSS images are identified by evaluating inter-band correlation between spectral channels. In vegetated areas, the green–red (bands 1–2) and NIR1–NIR2 (bands 3–4) reflectance values are typically highly correlated. For example, correlation coefficients above 0.9 are common in forested regions (Table 3). For comparison, the low-quality image in Figure 2 has a green-red correlation of 0.62 and a NIR1-NIR2 correlation of 0.82. TSDP flags and excludes MSS images where the correlation between either of these band pairs falls below 0.8, indicating poor data quality.

To improve the geometric accuracy of remaining Landsat MSS images, TSDP can optionally coregister them to the reference TM image. Co-registration vectors are computed using the displacement function in GEE (Google, 2023), which estimates pixel-wise shifts required to align each individual MSS image with the TM reference. Specifically, the NIR2 band of each MSS image is matched to the NIR band of the TM image. The near-infrared bands are preferred over visible bands for this operation because they typically provide higher image contrast and more distinct features, especially in vegetated and mountainous regions. The use of visible bands may lead to less accurate results due to lower contrast and increased sensitivity to atmospheric conditions. Once the displacement field is calculated, the MSS image is geometrically corrected by applying the estimated shifts to all MSS spectral bands.

Visual inspection indicated improved alignment among MSS scenes after co-registration, particularly along roads and forest clearings. However, quantitative assessment comparing three Landsat 5 MSS images with same-day Tier 1 Landsat 5 TM acquisitions using 77 ground control points (GCPs)[1] showed a relatively modest improvement. The RMSE decreased from 28.1 m before co-registration (consistent with the metadata value of 25.9 m) to 22.4 m after co-registration, corresponding to a ~ 20% reduction in positional error (~0.1 MSS pixel at 60 m resolution). Whether this improvement translates into better performance likely depends on the methods used for information extraction. In our tests, co-registration did not enhance forest disturbance detection using UC-Change 2.0, an updated version of the change detection algorithm described in Parshakov et al. (2025). One possible explanation is that UC-Change 2.0 is inherently robust to minor misalignments. Alternatively, spatial resampling during co-registration may have degraded the spatial detail of the input data due to interpolation, offsetting potential benefits.

3.6. Landsat MSS cloud masking

Cloud masks derived from the Quality Assessment (QA) bands of Landsat MSS imagery tend to be highly conservative, often leaving many cloud and cloud-shadow pixels unmasked. Therefore, TSDP applies an additional cloud-masking procedure based on the MSScvm algorithm by Braaten et al. (2015).

The method begins by converting MSS digital numbers to top-of-atmosphere (TOA) reflectance using the reflectance multiplication and addition coefficients, along with the solar zenith angle, all obtained from the image metadata. Clouds are then identified using a brightness test applied to the green band together with the normalized difference green–red index:

$$NDGR = (Red - Green)/(Red + Green), \tag{1}$$

where *Red* and *Green* are the TOA reflectance values for the corresponding spectral bands. Pixels are classified as clouds if either of two conditions is met:

(1) Green TOA reflectance exceeds 39% (MSS_bright_thres_green_ultimate), or (2) Green > 17.5% (MSS_bright_thres_green) and NDGR > 0.0.

Small isolated detections are removed using a connected-component sieve, and a spatial buffer is applied around the remaining cloud features to capture semi-transparent cloud edges.

The algorithm then separates water and shadowed slopes from true cloud shadows. Water is identified using a combination of NDVI and local slope, which helps distinguish water from other dark surfaces. Small artifacts are removed and a buffer is added around water boundaries.

A topographic correction (described in Section 3.10) is applied next to reduce terrain-induced shading. Cloud shadows are then detected in the NIR2 band using adaptive thresholds derived from image-level statistics. Only dark pixels within a geometrically plausible shadow zone—computed by projecting each cloud according to the solar zenith and azimuth angles for heights between 1 and 7 km—are retained as true shadows. This range captures typical low- and mid-level clouds, while excluding higher clouds such as cirrus, which are usually not detected in MSS data. Extending the projection beyond this range would therefore increase the likelihood of falsely identifying dark surfaces as cloud shadows (Braaten et al., 2015). This greatly reduces confusion with wetlands and other naturally dark land surfaces. A spatial buffer (240 m, or four MSS pixels) is applied around the final shadow detections.

The cleaned cloud and shadow layers are merged to produce a clear-view mask. This mask is then applied to the original raw DN image or the TOA image (if convert_MSS_to_TOA = True). If the parameter do_topo is set to True, TSDP performs a second SCS+C topographic correction on the cloud-masked TOA image (the initial correction is discarded because unmasked clouds degrade its quality). Thus, the outputs are either 8-bit raw DN MSS images or 16-bit TOA reflectance data with or without topographic correction.

TSDP implements a slightly modified version of the MSScvm method. It uses SCS+C instead of the Minnaert correction (advantages are discussed in Section 3.10) and applies more aggressive cloud-brightness thresholds—15% instead of 17.5% for MSS_bright_thres_green and 25% instead of 39% for MSS_bright_thres_green_ultimate—

as well as a larger default cloud buffer (240 m instead of 120 m). The original threshold values recommended by the developers of MSScvm were derived based on 12 Landsat scenes across the United States representing a wide range of land-cover and cloud types (Braaten et al., 2015). Those general-purpose thresholds are a sensible starting point, but we recommend users tune them for their specific application: select the lowest (i.e., most aggressive) thresholds that still avoid misclassifying objects of interest as clouds. TSDP's defaults were informed by visual inspection of multi-year time series over forested areas in Canada and were selected to balance sensitivity to thin clouds and cloud edges (reducing omission errors of cloud detection) with the need to avoid flagging bright forest disturbances (e.g., clearcuts) as clouds (reducing commission errors). For other use cases (e.g., agricultural or urban mapping), users should adjust cloud thresholds and buffer size based on the spectral separability between clouds and the objects of interest, preferably informed by region- and application-specific testing.

In addition, TSDP removes across-track striping before cloud masking. These stripes appear as bright and dark lines caused by corrupted pixel values in individual bands (Figure 2). To detect them, TSDP uses two normalized-difference indices, NDGR and the ratio (*NIR2* – *NIR1*) / (*NIR2* + *NIR1*), with valid values defined as −0.2 to 0.45 and −0.1 to 0.4, respectively. In forested regions of Canada, this approach works well because these band pairs are normally highly correlated, while striping introduces uncorrelated values that stand out. Adjustments may be needed for other regions or land-cover types.

Cloud masking remains challenging for Landsat MSS because the sensor lacks a blue, cirrus, or SWIR band—key spectral channels used by modern cloud-detection algorithms. The limited spectral resolution makes it difficult to reliably distinguish thin clouds from bright surfaces such as bare soil or recent clearcuts, which often have similar reflectance characteristics (see Figure 3). Therefore, when same-day Landsat TM imagery is available for Landsat 4 and 5 MSS acquisitions (where both MSS and TM sensors were mounted on the same satellites and acquired imagery simultaneously), TSDP applies the cloud and other masks derived from the TM data directly to the corresponding MSS images.

Finally, all Landsat 5 MSS images acquired in 2012–2013 lack valid NIR2 data (and lack simultaneously acquired TM data because the Landsat 5 TM sensor failed in 2011). For these scenes, TSDP instead uses the NIR1 band for cloud masking. NIR1 and NIR2 have similar reflectance over vegetated surfaces, although NIR1 is more affected by atmospheric scattering. This reduces contrast between cloud-shadowed and sunlit areas and may decrease shadow-detection reliability for these late-MSS scenes.

3.7. Sentinel-2 cloud masking parameters

TSDP uses the Cloud Score+ dataset available on GEE to mask clouds in Sentinel-2 imagery, which has been demonstrated to outperform other cloud masks for Sentinel-2 (Pasquarella et al., 2023). Cloud Score+ includes two quality assurance (QA) bands for each image: cs and cs_cdf (Google, n.d.-a). Both bands evaluate the usability of individual pixels with respect to surface visibility, scoring on a continuous scale from 0 (not clear or occluded) to 1 (clear or unobstructed). The cs band is based on spectral distance, measuring the similarity between the observed pixel and a theoretical clear reference, providing an instantaneous score of how similar a pixel is to a clear observation. On the other hand, the cs_cdf band

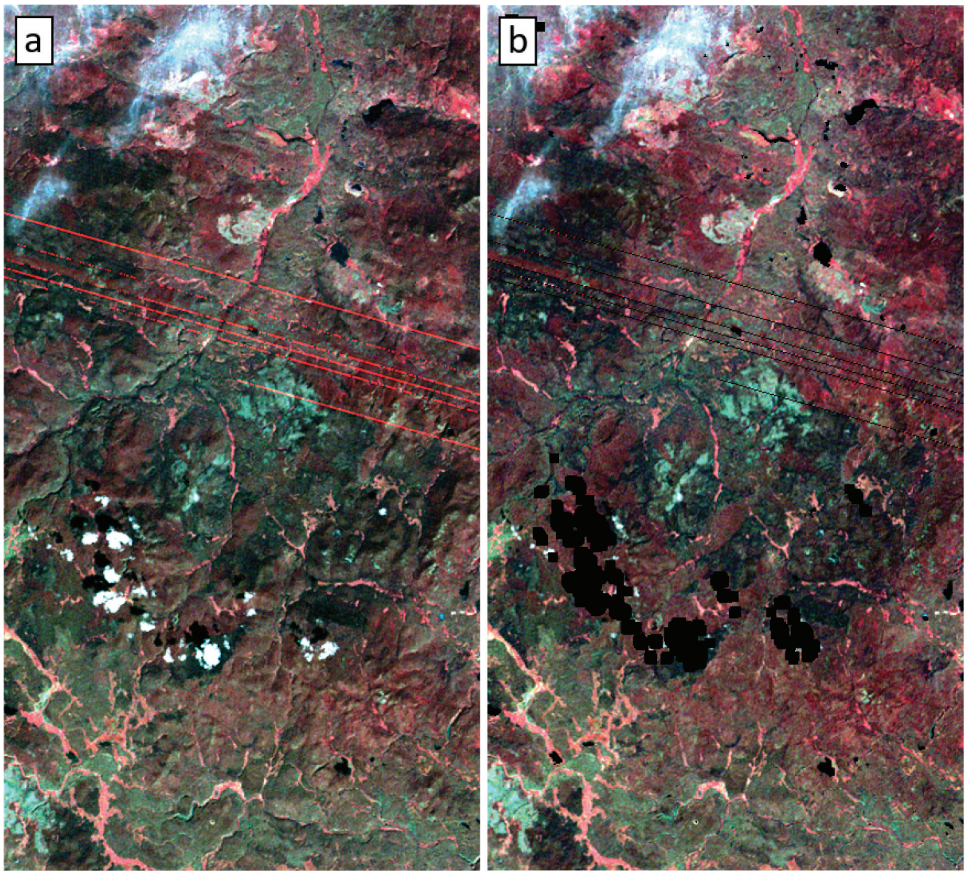


Figure 3. Landsat 1 MSS image acquired on 1972-08-19 (center coordinates: 52°58'N, 123°53'W; RGB = NIR2-Red-green), shown before (a) and after (b) cloud masking and topographic correction. Cloud masking was successful in the southern (bottom) part of the image subset but failed in the northern (top) portion. Thin clouds in the top half have relatively low green toa reflectance—similar to bare ground and recent clearcuts—and fall below the masking threshold (green TOA > 15%). Also note the flatter appearance of the topographically corrected image, where illumination differences caused by terrain shadows and slope orientation have been reduced (see Section 3.10).

reflects the likelihood that a pixel is clear, estimated by considering the cumulative distribution of cloud scores for the same location over time, offering a temporal context for cloud presence. The choice of Cloud Score+ band influences cloud masking: the cs band is more responsive to haze and cloud edges, providing stricter identification of clear pixels, while the cs_cdf band is less sensitive to subtle spectral variations and terrain shadows, reducing false cloud detections (Pasquarella, 2023). The Cloud Score+ Collection Explorer (Google, n.d.-b) is an interactive tool that allows users to quickly identify the most suitable Cloud Score+ band and threshold for a given study area.

By default, TSDP uses the cs band with a threshold of cs > 0.8 applied for more aggressive cloud masking. This higher threshold (compared to ~0.6 recommended by the developers) prioritizes quality over quantity, as the abundance of post-2017 imagery

generally ensures that the quantity of Sentinel-2 data is not a limiting factor in time series analyses.

3.8. Snow masking thresholds

Snow masking in optical satellite imagery is an important preprocessing step for applications such as change detection, land-cover classification, and time-series analysis (Friedl et al., 2022; Lang et al., 2023; Shang et al., 2022). USGS provides snow and ice masks within the QA bands of Landsat data, facilitating automated masking for Landsat 4–9 Collection 2 products. However, Landsat MSS data lack built-in snow masks, and Sentinel-2 snow classifications often exhibit high omission errors (based on visual assessment), necessitating custom masking approaches.

TSDP addresses these limitations by leveraging the spectral characteristics of snow, which has high reflectance in the visible bands and low reflectance in the shortwave infrared (SWIR) bands (Kulkarni et al., 2002). Users can apply the Normalized Difference Snow Index (NDSI), defined as (Green − SWIR1) / (Green + SWIR1), or set individual thresholds for the green and SWIR1 bands to identify snow-covered pixels. The NDSI is widely used, with a common threshold of NDSI > 0.4 for atmospherically corrected imagery (Dozier, 1989; Poussin et al., 2023; Zhang et al., 2019). As a spectral index, NDSI is less sensitive to topographic illumination effects compared to individual bands (Negi et al., 2010). However, its performance can be influenced by bidirectional reflectance distribution function (BRDF) effects, snow physical properties, atmospheric conditions (especially in top-of-atmosphere data where atmospheric scattering affects visible reflectance more than SWIR), and vegetation cover (e.g., forest type) (Wang et al., 2022). Consequently, a universal threshold may not be optimal across different geographic regions. For instance, in central British Columbia, we found a lower threshold of NDSI = 0.0 to be more effective for snow masking.

Alternatively, applying thresholds to individual bands allows for more intuitive adjustments tailored to specific study areas or regions. In coniferous forests of Canada, we found that setting a green TOA reflectance threshold > 20% and a SWIR1 TOA reflectance threshold < 10% effectively masked snow while preserving clearcuts and dense forest areas. After snow, clearcuts were typically the brightest objects in visible bands, while dense conifer forests had the lowest SWIR reflectance (after water and snow). Applying both thresholds simultaneously (green > 20% and SWIR1 < 10%) allowed for effective separation of snow from land-cover types critical for forest change detection.

Landsat MSS data lack SWIR bands, precluding the use of NDSI or SWIR-based thresholds. For MSS, TSDP employs the green band for snow, cloud, and smoke masking. Notably, TSDP does not attempt to distinguish among these bright features and instead treats all bright areas as potential cloud cover. Consequently, it may also mask nearby dark pixels as cloud shadows (see Section 3.6), regardless of whether the adjacent bright object is snow, smoke, or actual cloud, which may lead to over-masking.

To improve snow masking, TSDP applies a spatial buffer (e.g., 240 meters by default) around identified snow pixels. This dilation accounts for pixels that are likely to also contain snow but do not meet the detection thresholds.

Applying a time series approach can further improve snow masks generated through threshold-based single-date masking. For instance, Zhu and Woodcock (2014) detected

additional snow and cloud pixels overlooked by single-date masking by comparing observed Landsat reflectance with predictions from models simulating seasonal reflectance changes. This capability is not yet implemented in the current version of TSDP. Recent work with generative AI—though currently aimed at replacing snow-covered pixels with synthetic snow-free ones rather than masking—also shows promise for improving snow detection, since these models learn to distinguish snow from other surfaces (Oluwadare et al., 2024; Zhang et al., 2023).

3.9. Wildfire smoke masking thresholds

Similar to snow and ice, smoke from wildfires exhibits relatively high visible reflectance and low SWIR reflectance. Therefore, thresholds based on NDSI, the green band, and the SWIR1 band can also mask some smoke pixels. However, smoke often appears less bright than snow in the visible spectrum and may be more variable in optical thickness, making dedicated smoke masking thresholds beneficial. In Landsat 4–7 data, using the blue band rather than the green band can improve smoke detection because smoke tends to scatter shorter wavelengths more strongly, resulting in greater contrast between smoke and the underlying surface in the blue band. Thus, applying a blue-band reflectance threshold (in addition to the SWIR1 threshold) can be more effective for masking smoke in these datasets.

For Landsat 8–9 and Sentinel-2, the availability of aerosol-specific bands further enhances smoke detection. The coastal aerosol band (band 1) in Landsat 8–9 and the similar band in Sentinel-2 (band 1) capture even shorter wavelengths than the blue band, increasing sensitivity to fine atmospheric particles such as smoke. Thresholds applied to these aerosol bands can better distinguish smoke from land surfaces, particularly when used alongside SWIR bands to exclude bright surfaces like sand or urban areas. TSDP automatically uses the aerosol band instead of the blue band for masking smoke in Landsat 8 and 9 and Sentinel-2 data (Figure 4). Some smoke is masked together with clouds in Sentinel-2 images when using the Cloud Score+ dataset.

3.10. Topographic illumination correction

Topographic variation can introduce significant artifacts into satellite data analysis, particularly in mountainous regions where terrain orientation relative to solar illumination causes systematic biases in pixel values (Parshakov et al., 2025). To mitigate these effects, the TSDP script can optionally implement a topographic illumination correction based on the Sun-Canopy-Sensor plus C (SCS+C) correction method (Soenen et al., 2005). This method relies on a Digital Elevation Model (DEM) to derive the local slope and aspect of each pixel, which are critical inputs for accurately modeling illumination conditions across complex terrain.

The SCS+C method modifies the standard Sun-Canopy-Sensor (SCS) correction by introducing an additional empirical parameter (C) that stabilizes the correction, especially in areas with low illumination, thereby improving reflectance normalization across slopes with varying aspects and inclinations. The correction uses a DEM to compute the illumination factor, defined as the sum of the cosine of the terrain slope (from the DEM) multiplied by the cosine of the solar zenith angle, and the sine

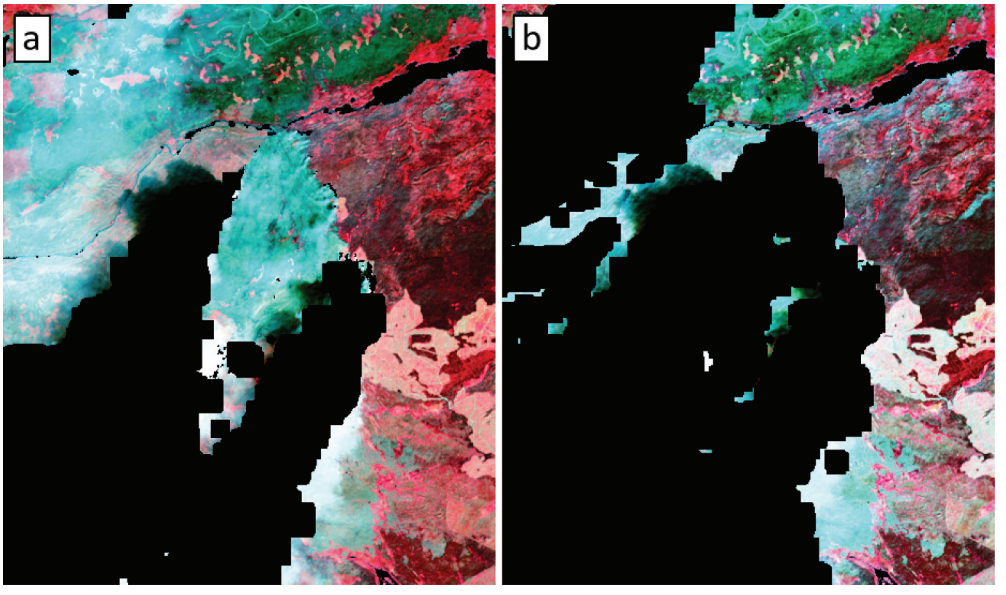


Figure 4. Landsat 8 image acquired on 2017-08-04 (center coordinates: 53°10′N, 124°2′W; RGB = NIR-red-green), showing wildfire smoke masking results: (a) using only the quality assessment band and (b) using additional thresholds applied to the aerosol band (band 1) and SWIR1 (band 6). The latter approach shows improved detection and masking of smoke.

of the slope multiplied by the sine of the solar zenith angle times the cosine of the relative azimuth between the sun and the terrain aspect. In other words, this factor describes the angle between incoming solar radiation and the terrain surface. A linear regression between the observed reflectance and the illumination factor is then fitted, and the ratio of the regression intercept to the slope (the C parameter) is used to adjust the correction, reducing overcorrection in shadowed areas by emulating the effect of diffuse sky illumination in each particular spectral band (Soenen et al., 2005). The following formula represents the SCS+C topographic correction method:

$$BV_{\text{corrected}} = BV \times \frac{\cos(\alpha) \times \cos(\theta) + C}{\cos(i) + C}, \tag{2}$$

where BV and $BV_{\text{corrected}}$ are the original and topographically corrected pixel brightness values; α is the terrain slope; θ is the solar zenith angle at the time of image acquisition; i is the illumination (incidence) angle between the sun and terrain normal; C is the empirical correction factor from linear regression.

We modified the SCS+C topographic correction by setting a minimum threshold of zero for the illumination factor, cos(i), after calculating C and before using the illumination factor in Equation 2. The illumination factor can theoretically range from −1 to 1 but is only physically meaningful for values between 0 and 1. On steep slopes facing away from the sun or under low solar elevation, when no direct sunlight reaches the surface (i.e., $i \geq 90°$), the illumination factor can approach zero or become negative. Without this modification, the SCS+C algorithm may produce negative pixel values under such conditions.

In the TSDP implementation, the SCS+C correction is applied band-by-band across the satellite image using parallelized computation to efficiently handle large datasets. Only valid pixels (i.e., non-masked and non-zero values) are included in the regression and correction steps to avoid introducing artifacts. The correction can be applied using any DEM available in GEE, and the user can specify their preferred DEM source.

This approach was found particularly suitable for medium-resolution sensors such as Landsat, where simpler terrain correction methods (e.g., Cosine or Minnaert corrections) based solely on solar geometry often result in under- or overestimation of reflectance on steep terrain (Wu et al., 2016). By dynamically deriving the C parameter for each image and each spectral band and using DEM-based terrain geometry, the SCS+C method provides an image-specific correction that improves the consistency of pixel values across varying topography.

Figure 5 compares a Sentinel-2 image without topographic correction, with correction applied by TSDP using the Copernicus DEM GLO-30 and NASADEM (these DEMs are described in Section 2.2), and with the standard topographic correction provided by ESA for Level 2A (SR) imagery. Based on visual assessment, the Copernicus DEM GLO-30 yielded the best results for small-scale terrain features, such as small ridges and valleys, while NASADEM was slightly more effective at flattening broad ridges and deep valleys. It is worth noting, however, that both of these DEMs have coarser spatial resolution than Sentinel-2 imagery, resulting in finer terrain variability being underrepresented in either case. Nevertheless, in contrast to the SCS+C topographic correction results produced by TSDP, ESA's correction failed to adequately compensate for illumination effects in the provided example, often producing pixel values that were unrealistically bright or dark.

It is important to note that topographic correction is not a universal solution, and its application in the TSDP algorithm should be weighed carefully in light of potential trade-offs. While SCS+C can substantially reduce the variability in pixel values caused by topographic effects (Hurni et al., 2019) and produce visually compelling results, it may also introduce new errors by altering the relationship among spectral bands (Hurni et al., 2019). In preliminary tests in mountainous areas, UC-Change 2.0 showed no overall improvement in forest disturbance detection with topographically corrected data across all sensors tested (Landsat MSS, TM, and Sentinel-2). For Landsat MSS, detection of clearcuts on steep slopes (>30°) improved by 12.0%, but overall accuracy decreased by 2.2%. For Landsat TM and Sentinel-2, topographic correction resulted in no benefit or a slight reduction in detection accuracy, even on steep slopes (up to 3.0% reduction). It also had minimal effect on false positive rates, reducing them only slightly: from 0.64% to 0.61% for MSS, 0.05% to 0.04% for TM, and 0.05% to 0.03% for Sentinel-2. These findings are specific to UC-Change 2.0, and readers should not generalize them to other time-series analysis methods, some of which may be more sensitive to topographic effects.

Another limitation is that topographic correction cannot compensate for increased image noise on steep, sun-opposing slopes—particularly in older sensors with lower signal-to-noise ratios (e.g., Landsat MSS or Landsat 7 ETM+ panchromatic). For these sensors, reduced accuracy on such slopes should be expected, regardless of the correction method or DEM used.

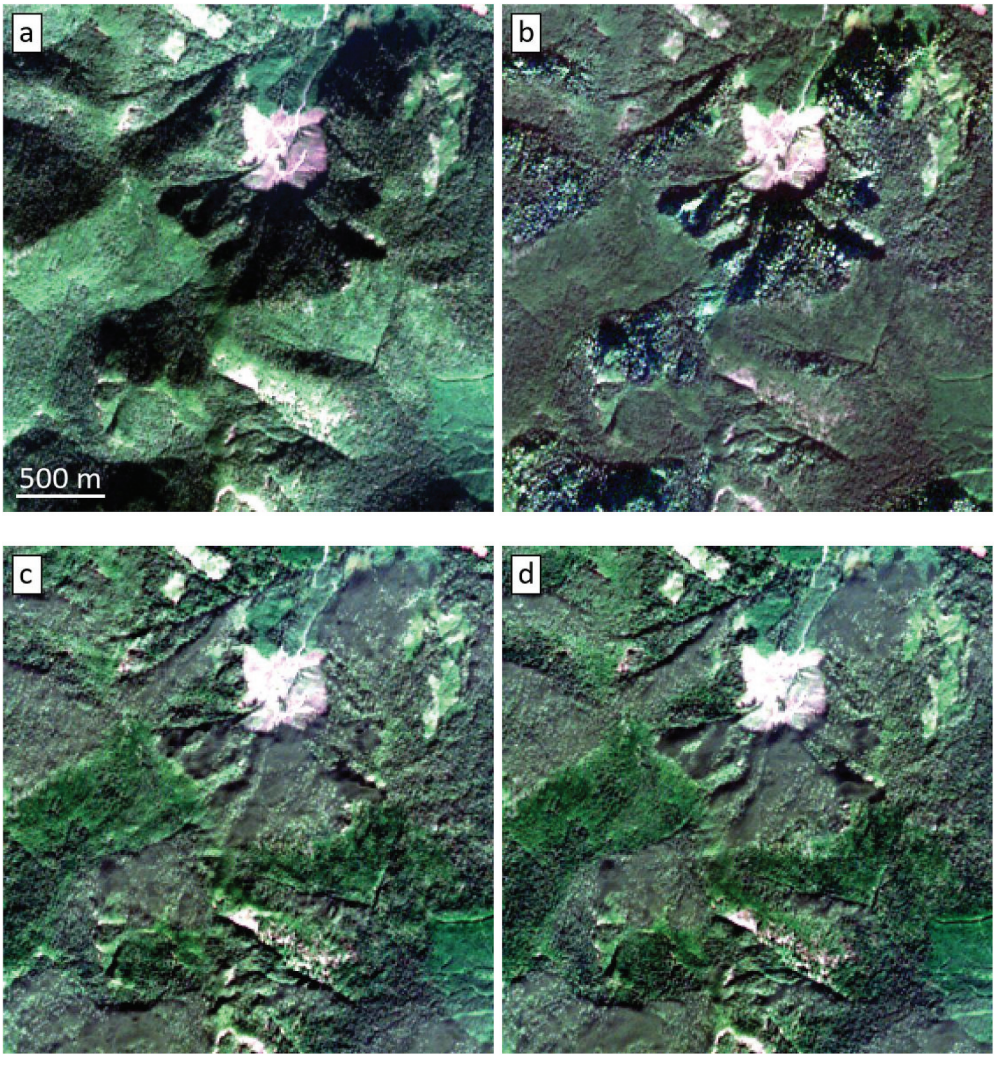


Figure 5. Sentinel-2 image from 2023-09-20 (center: 48°45′N, 124°38′W), shown in true color without topographic correction (a); with ESA Level-2A correction (b; processing baseline version 5.10); with TSDP correction using NASADEM (c); and using Copernicus DEM GLO-30 (d). This is one of the more challenging examples: a low solar zenith angle of 49° combined with extreme topography. Under less extreme conditions, topographic correction typically produces better results. Figure 3 provides another example.

4. TSDP output

By default, the algorithm downloads only images with ≤30% cloud cover, except for particularly cloudy years, identified as when the 30% threshold results in <95% yearly coverage of the study area. For those years, the threshold is increased (up to 80% cloud cover) until the images provide 95% coverage for each sensor group (e.g., Landsat 4 TM–9 OLI is one group and Sentinel-2 MSI is another). For the years 1999–2012, this rule is first applied to both Landsat 5 TM and Landsat 7 ETM+

together, and then, optionally, only to Landsat 7 ETM+, to ensure 90% yearly coverage by the 15 m panchromatic ETM+ data.

Images are downloaded directly to the user's computer memory, eliminating the need for Google Drive storage required by alternative methods. Each downloaded image is then saved in a folder named after the user-defined MGRS tile as two NumPy arrays: one containing pixel values and the other containing pixel locations within the image (e.g., "C:/TSDP/10UCV/2022-08-16_L8_data.npy" and "C:/TSDP/10UCV/2022-08-16_L8_mask.npy"). This form of lossless compression significantly reduces storage requirements, particularly for images with a high proportion of masked pixels (e.g., due to cloud or water masking).

Image downloading is highly parallelized, with the algorithm retrieving all spectral bands simultaneously. When downloading from GEE, TSDP also subdivides each band into multiple subtiles (up to 16 per band) to comply with GEE's download size limit of 48 MB per image chip. After they are downloaded, image chips are automatically mosaicked back together.

Additionally, TSDP provides the option to convert the output into multiband GeoTIFF files accompanied by ENVI header files containing band names and wavelengths (e.g., "C:/TSDP/10UCV_tif/2022-08-16_L8.tif" and "C:/TSDP/10UCV_tif /2022-08-16_L8.hdr"). This format ensures compatibility with popular GIS and remote sensing software packages, such as ENVI, ArcGIS, and QGIS (Figure 6).

All downloaded data have the UTM projection of the corresponding MGRS tile (for example, all images for MGRS tile 10UDD use the UTM 10 projection). As a result, TSDP does not reproject Sentinel-2 imagery and most Landsat data. However, some Landsat images spanning adjacent UTM zones require reprojection because USGS does not provide separate copies of an image for each UTM zone; instead, each Landsat Level-1 and Level-2 product is provided in the UTM zone corresponding to the center of the scene. Consequently, TSDP automatically reprojects Landsat data as needed when creating subsets for the selected MGRS tile using the nearest neighbour resampling method. The effect of this resampling on data quality has not yet been assessed.

While the downloaded images from different sensors have the same metric dimensions (e.g., all images are 102,960 m wide and 102,960 m long if a 2,960-m tile buffer is used), they cover slightly different areas on the ground due to pixel grid misalignment, as discussed in Section 2.1. Consequently, the northwest corner of Landsat panchromatic images is shifted 7.5 meters south and 7.5 meters east relative to Sentinel-2 images. For Landsat 4–9 multispectral data, this shift is 15 meters, and for Landsat MSS data, it is 30 meters.

4.1. Storage requirements

The storage requirements for data downloaded using TSDP depend on several user-defined parameters, including the selected time period, day-of-year range, spectral bands, and data preprocessing level (e.g., DN vs. SR, and whether topographic correction or MSS-to-TM image coregistration is applied). For instance, retrieving Landsat 1–9 (including panchromatic data), Sentinel-2, and GEDI data for MGRS tile 10UCV over the period 1972–2023 using the default settings (i.e., no topographic correction or coregistration, and the lowest processing level—raw DN for Landsat 1–9) resulted in 49.3 GB of data

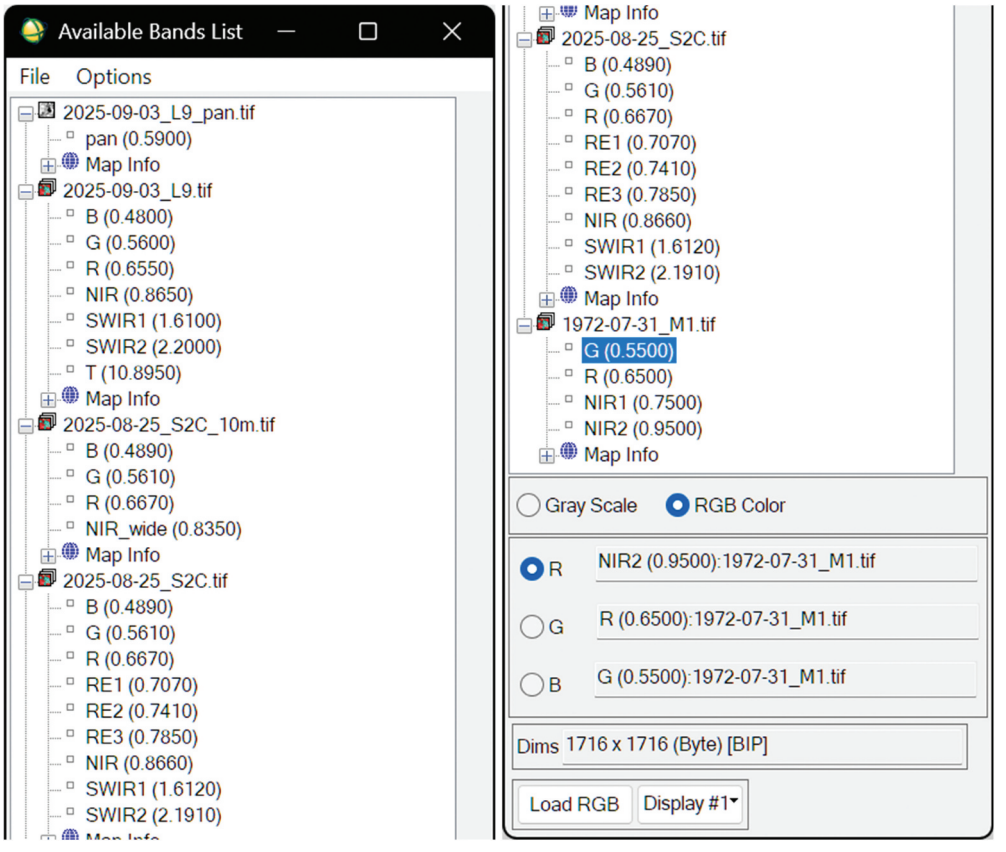


Figure 6. Example of multiband GeoTIFF files generated by TSDP and opened in ENVI Classic. Each file includes georeferencing information, along with human-readable band names and wavelengths for ease of use.

when saved as NumPy arrays. Storage requirements increased when downloading 16-bit Landsat 1–7 images that had been topographically and/or atmospherically corrected—rather than the original 8-bit DN values—resulting in 56.2 GB of NumPy files for the same set of data.

Saving as compressed GeoTIFF files typically requires slightly less storage because a different, more storage-efficient lossless compression method is used, compared to the custom compression method applied to NumPy arrays, which was described in Section 4. TSDP uses the Lempel–Ziv–Welch (LZW) algorithm to compress GeoTIFFs—a dictionary-based technique that replaces repeated byte sequences with shorter codes while preserving all original information (Welch, 1984). This compression method is widely supported by remote sensing and GIS software but significantly increases the time required to open files compared to uncompressed GeoTIFFs. Therefore, saving as uncompressed GeoTIFF is the default option (controlled by the parameter compress_tif).

Most users are likely to require only one of the two formats: either NumPy files for Python-based processing, or GeoTIFFs for use in existing GIS and remote sensing software

such as ENVI, ArcGIS, or QGIS. The TSDP algorithm itself, implemented as a single Jupyter Notebook, is very lightweight, requiring only about a quarter of a megabyte of storage.

Like file size, download speed is similarly influenced by user choices, particularly the complexity of preprocessing applied within GEE prior to export. In our tests using a 100 Mbps connection, export times varied considerably: applying minimal preprocessing to Landsat TM images required approximately 20–30 seconds per scene, whereas the more extensive preprocessing applied to Landsat MSS imagery (artifact filtering, coregistration) increased export time to approximately a minute per image, despite MSS files being much smaller.

5. Intended use and limitations

TSDP is intended for academic research and development purposes and is best suited for downloading moderate volumes of satellite data (one MGRS tile at a time)—such as those required for algorithm prototyping or localized case studies. Due to limitations in GEE, TSDP cannot be used to download very large volumes of data, as such usage may cause repeated failures, service timeouts, or unresponsiveness in the GEE environment. Users are encouraged to familiarize themselves with Google Earth Engine's Terms of Service for additional guidance on platform usage restrictions (Google, n.d.-c).

Once the user's algorithm is mature and ready for large-scale or operational deployment, we recommend transitioning to a fully cloud-based implementation (e.g., within GEE itself) or using alternative bulk download methods directly from data providers, such as the Copernicus Open Access Hub or the USGS EarthExplorer. TSDP currently supports direct downloads from the Copernicus Hub (for Sentinel-2) and NASA's Earthdata platform (for HLS data), but not for Landsat Collection 2. A future update will add the option to download Landsat Collection 2 directly from the USGS to improve scalability and reduce reliance on GEE infrastructure.

TSDP was originally developed to generate input data for the UC-Change 2.0 algorithm (to be published and released as open source shortly after TSDP), and some parameter tuning or code modifications may be required to optimize it for other applications. UC-Change is a forest change detection technique designed to ingest data from Landsat MSS and newer sensors in their native spatial and spectral resolutions. Consequently, TSDP does not perform spatial resampling (with rare exceptions; see Section 4) or radiometric normalization. While this design choice preserves sensor-specific information, it means that TSDP outputs are not directly compatible with pixel-based time-series image analysis methods, such as LandTrendr and CCDC, which require a consistent spatial and spectral resolution. Compatibility is only possible if data from each sensor are processed separately.

That said, TSDP is fully open source, and users can readily adapt it to alternative workflows. For example, with minor code modifications, TSDP could resample Sentinel-2 imagery to Landsat's 30 m pixel grid during data download. In addition, TSDP already supports downloading 30 m Harmonized Landsat–Sentinel (HLS) data, which are suitable for pixel-based change detection approaches, although these data are only available from 2013 onward.

Another consideration when using TSDP for applications outside the UC-Change framework is that TSDP uses relatively aggressive default parameter settings for

masking cloud, snow, or smoke pixels to prioritize data quality (e.g., minimizing residual cloud contamination) over quantity. For example, TSDP applies a 240 m cloud buffer by default, compared to the 120 m buffer recommended for Landsat MSS (Braaten et al., 2015) and the 100 m buffer suggested for Sentinel-2 (Zekoll et al., 2021). While stricter thresholds and wider buffers reduce residual cloud, smoke, or snow contamination, they also increase the probability of masking valid, clear pixels, thereby reducing the temporal density of observations for some locations.

This parameter choice can reduce commission errors without adversely affecting the producer's accuracy of forest change detection by UC-Change, which is able to operate effectively on both sparse and dense time series. However, it is less suitable for methods such as CCDC or LandTrendr, which are more sensitive to data availability (Zhu & Woodcock, 2014; Zhang et al., 2021; Parshakov et al., 2025. Users who intend to download data for applications other than UC-Change may therefore benefit from relaxing these parameter settings to better balance data quantity and quality for their specific analysis. Because TSDP relies mostly on established preprocessing tools, readers are referred to the original publications and documentation of those tools for guidance on parameter optimization. Nonetheless, sensitivity analysis is recommended, as no single parameter set can be expected to perform optimally across all applications (e.g., crop classification versus forest disturbance detection) and all geographic regions (e.g., boreal versus tropical forests). For each group of optical sensors (Landsat 1–5 MSS, Landsat 4 TM–9 OLI, and Sentinel-2), TSDP generates a timelapse video (e.g., "Landsat MSS timelapse. mp4") that allows rapid visual inspection of all downloaded images, facilitating manual threshold adjustment or the removal of low-quality images. This is particularly helpful for adjusting fixed user-defined spectral thresholds, such as those used for snow and smoke masking, as well as Landsat MSS cloud masking.

TSDP is intentionally designed around the MGRS tiling system to enable parallel, reproducible downloading and preprocessing of multi-sensor data using a minimal set of easily installable Python dependencies. Consequently, the tool currently operates exclusively on MGRS tiles and does not support arbitrary user-defined areas of interest (AOIs). This design choice reflects the native tiling structure of Sentinel-2 products and ensures consistent spatial coverage across sensors with minimal reprojection or mosaicking. While this approach may result in downloading more data than required for small or irregular study areas, users can subsequently clip the tiled outputs to their AOI using standard GIS or RS software or batch-processing scripts—an extra step we acknowledge as a current limitation. For AOIs that span multiple MGRS zones, however, the required cross-tile mosaicking and reprojection logic falls outside the current scope of the tool.

6. Conclusion

The preparation of multi-decadal, multi-sensor satellite image time series remains one of the most time-consuming components of remote sensing research. Despite the availability of batch download utilities and analysis-ready products, researchers still encounter fragmented workflows, inconsistent preprocessing routines, and limited support for the full Landsat archive. TSDP addresses these gaps by consolidating data acquisition and preprocessing within a single framework. It combines access to multiple data providers,

supports the complete Landsat record from Landsat 1 MSS onward, and integrates a broader set of preprocessing routines than existing downloaders.

TSDP leverages multi-platform data acquisition to combine the strengths of individual services. For example, it enables retrieval of the full Landsat archive and advanced cloud masking for Sentinel-2 through GEE, while also allowing access to Sentinel-2 imagery from the Copernicus Hub with the most recent ESA processing baselines. By coordinating these sources within one workflow, TSDP reduces the need for manual cross-platform data management and ensures the consistent quality of Sentinel-2 data across time.

The workflow applies a consistent preprocessing framework—image selection, cloud, snow, smoke, agriculture, and water masking, topographic correction, and coregistration—across sensors spanning more than five decades, from Landsat MSS to Landsat OLI and Sentinel-2. This produces standardized time series from 1972 to the present, ensuring uniform processing and compatibility across sensors for downstream analyses.

Unlike harmonized analysis-ready datasets such as HLS, which resample imagery to a common pixel grid to facilitate pixel-based methods, TSDP maintains native spatial and spectral resolutions. Retaining Sentinel-2's 10 m bands and Landsat's 15 m panchromatic data in particular enables emerging multi-resolution approaches to exploit fine spatial detail where it exists, rather than sacrificing it for strict cross-sensor uniformity.

By lowering the practical barriers to constructing long time series, TSDP enables researchers to allocate effort toward methodological development rather than data logistics. Its design aligns with the growing demand for multi-resolution information extraction approaches that leverage sensor-specific characteristics instead of suppressing them through harmonization. In this way, TSDP contributes a structured, reproducible data-handling framework that supports continued innovation in remote sensing research.

7. Future work

This paper describes TSDP version 1.3.0, which forms the foundation for several planned enhancements to expand functionality, sensor coverage, and usability. First, we aim to extend support to additional satellite missions, including ESA's Biomass (P-band SAR), NASA's ICESat-1 and ICESat-2 (spaceborne LiDAR), and other optical and SAR missions from public sources such as SPOT and Sentinel-1. These additions would allow users to integrate a broader range of multisensor data for more comprehensive ecosystem monitoring.

The tiling system could be optimized to reduce storage and processing demands without compromising spatial completeness. At the boundaries between adjacent UTM zones, tile overlap can become substantial—particularly at higher latitudes, where some tiles may nearly duplicate one another in spatial coverage (Bauer-Marschallinger & Falkner, 2023). A future version of TSDP will address this by incorporating an option to minimize overlap in these areas.

Landsat MSS data preprocessing could be improved in a number of ways. First, spectral thresholds used for cloud and snow masking could potentially be automatically adjusted on a per-tile basis using the high-quality cloud and snow masks available from same-day Landsat TM images, since the Landsat 4 and 5 satellites carried both MSS and TM sensors acquiring imagery simultaneously. These adjusted thresholds could then be applied to Landsat MSS images that lack accompanying TM imagery (e.g., all Landsat 1–3 MSS, much

of Landsat 4 MSS, and Landsat 5 MSS after the TM failure in 2012). In the current implementation of TSDP, TM-derived cloud and snow masks are applied directly to the corresponding same-day MSS images, but are not used to calibrate MSS cloud-masking thresholds.

Second, we plan to test a time-series-based cloud masking approach for Landsat MSS, leveraging techniques such as UC-Change to identify persistent features and remove transient artifacts like clouds and smoke. This can potentially improve cloud detection performance compared to the current single-image, threshold-based method—especially in the absence of dedicated cloud-sensitive bands.

We also plan to test alternative image coregistration methods for Landsat MSS data, such as the AROSICS algorithm (Scheffler et al., 2017). The current approach—using the displacement function in GEE—tends to blur fine spatial detail and reduce image sharpness, which can affect the quality of downstream analyses such as change detection. In contrast, AROSICS applies subpixel cross-correlation and distortion modeling to achieve accurate image alignment without excessive resampling, which could help preserve the native spatial detail of Landsat MSS imagery.

In addition, we intend to address incorrect georeferencing in some MSS images, which currently results in image rejection. By automatically detecting and correcting spatial misalignments, TSDP could retain more of this valuable historical data.

Another priority is to develop a standalone application with a graphical user interface (GUI). This will enhance accessibility for users unfamiliar with Python or coding environments, particularly students and domain scientists.

Finally, TSDP is fully open source and designed for user modification and secondary development. The entire workflow is implemented in a single, well-documented Jupyter Notebook, with preprocessing steps organized into clearly labeled sections. This structure allows users to easily inspect, modify, or extend individual components of the pipeline. Researchers can, for example, add new preprocessing routines, modify existing masking or filtering steps, or incorporate additional cloud-based (GEE) or local Python processing to generate secondary products. As a result, TSDP naturally supports extensibility and customization, even for users with limited experience in software development.

Note

1. GCPs were automatically generated using the Förstner operator in ENVI, which produced more reliable and repeatable tie points than the Moravec operator (the only other option available) in the coarse-resolution, low-contrast scenes with cross-sensor radiometric differences (MSS vs TM), due to its improved image matching capabilities (ENVI documentation).

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Author contributions

CRediT: **Ilia Parshakov:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing; **Dongmei Chen:** Funding acquisition, Resources, Supervision.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Notes on contributors



Ilia Parshakov conducted this work as a postdoctoral researcher in the Laboratory of Geographic Information and Spatial Analysis (LaGISA) at Queen's University. His work focuses on developing multisource, multi-resolution time-series algorithms for detecting forest disturbances using large satellite archives, including Landsat 1–9, Sentinel-2, and spaceborne LiDAR (GEDI, ICESat-2). He has over a decade of experience in remote sensing research, image preprocessing, and automated large-scale analysis workflows. His recent work includes developing the UC-Change algorithm for long-term (1972–present) forest disturbance mapping and methods for leveraging spaceborne LiDAR to improve biomass estimation.



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AI disclosure statement

This technical note was prepared with assistance from the generative AI tool ChatGPT (OpenAI, GPT v4.0 to v5.1, 2025). The tool was used to improve the clarity and readability of the manuscript and to help refine the wording of several sections. ChatGPT was also used for minor code optimizations in the TSDP Jupyter Notebook. All outputs generated by ChatGPT were reviewed, verified, and edited by the authors to ensure accuracy and appropriateness.

Data availability statement

The Time-Series Data Downloader (TSDP) code is open access and available on GitHub at https://github.com/parshakov-code/time-series-data-downloader. A sample of TSDP outputs (downloaded

imagery) is available in the same GitHub repository. All satellite datasets used or retrieved through TSDP are freely and openly available from their respective providers: Landsat data from Google Earth Engine and NASA Earthdata, and Sentinel-2 data from the Copernicus Data Space Ecosystem. TSDP only automates access to these publicly accessible archives and does not rely on any proprietary data.

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